USER INTERFACE DESIGN

EXPERIMENT – 2

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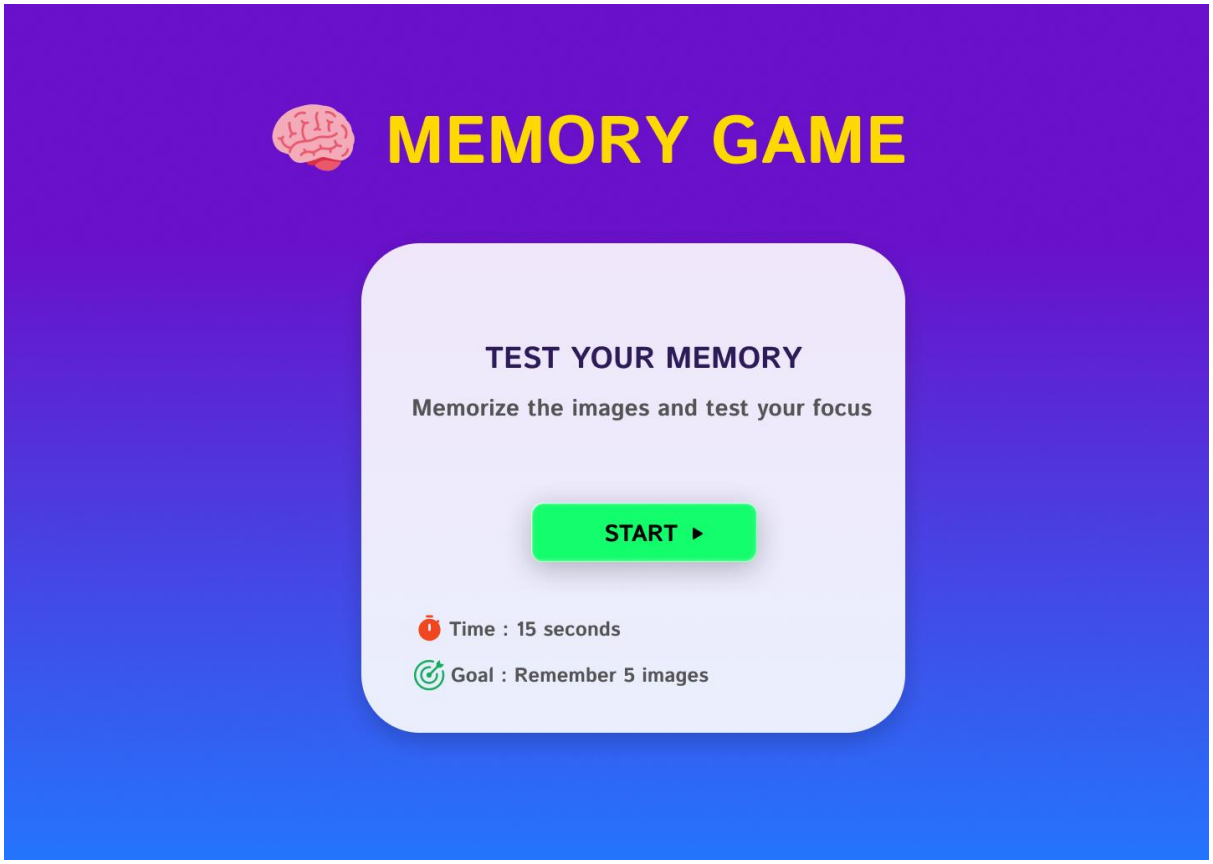
Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INSTRUCTION SCREEN



Purpose of the Screen

This screen introduces the user to the Memory Game and explains the task clearly before starting. It prepares the user mentally for the upcoming memory challenge.

Key UI Elements

- **Title: “MEMORY GAME”**
  Clearly indicates the purpose of the application.
- **Instruction Card**
  - “Test Your Memory”
  - “Memorize the images and test your focus”
  - Simple, readable content reduces cognitive load.
- **Start Button**
  - Highlighted in green to attract attention.
  - Indicates the beginning of the memory task.
  - Acts as the main call-to-action.
- **Time & Goal Info**
  - Time: 15 seconds
  - Goal: Remember 5 images
    This sets expectations before the game starts.

UX & Cognitive Benefit

- Instructions are short and direct.
- Visual hierarchy helps users understand what to do quickly.
- The clean layout avoids distraction and improves focus.

FRAME 2: CHUNKING PHASE (MEMORIZATION SCREEN)

Purpose of the Screen

This is the **visual encoding phase**, where users observe and memorize a set of images within a limited time.

Key UI Elements

- **Heading:** “SELECT THE IMAGES YOU REMEMBER”
- **Grid of Fruit Images**
  - Fruits are arranged neatly in rows.
  - Images are visually distinct (apple, banana, kiwi, pineapple, etc.)
  - Helps users form mental groups (chunking).
- **Timer / Progress Indicator**
  - Shows limited viewing time.
  - Creates urgency and forces quick memory processing.

Chunking Concept Used

Users naturally group images by:

- Fruit type
- Color
- Shape
- Familiarity

This improves short-term memory retention and makes recall easier.

UX Benefit

- Prevents overload by grouping items visually.
- Improves recall accuracy.
- Engages the user through visual learning.

FRAME 3: IMAGE SELECTION & VALIDATION SCREEN

Purpose of This Screen

This screen tests the user's memory by allowing them to select the images they remember from the previous screen.

Unlike traditional quiz systems, **there is no submit button**.
The system evaluates the user's selection **instantly**.

User Interaction Flow

- The user clicks on the **small circular button below each image**.
- Each click is checked immediately by the system.
- There are two possible outcomes:

Correct Selection Behaviour

- When the user selects a **correct image**:
  - The circular indicator turns **green**
  - The image is marked as correctly remembered
- The user can continue selecting more images
- If **all correct images are selected without mistakes**:
  - The system shows a **success feedback screen**
  - Message displayed:
    "Excellent Memory!"

✓ This creates positive reinforcement
✓ Encourages accuracy and attention
✓ No unnecessary confirmation steps

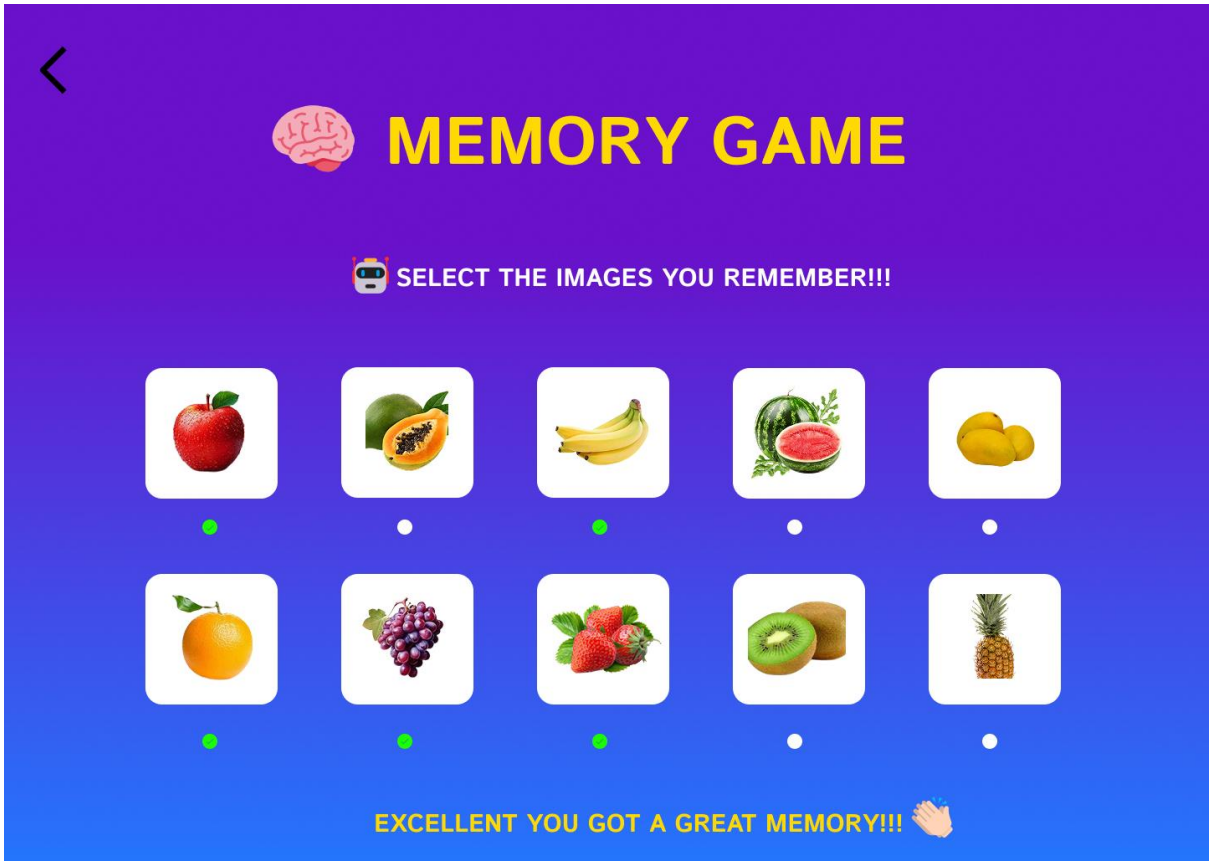
Wrong Selection Behaviour

- If the user selects **any wrong image**:
    - The selected wrong image turns **red**
    - All correct images are highlighted in **green**
    - The mistake is shown immediately

There is **no second chance** once a wrong image is clicked.

FRAME 4: RESULT / FEEDBACK SCREEN

If User Selects All Correct Images



- Feedback Message:

“Excellent You Got A Memory!”

- Visual Feedback:
    - All correct images highlighted in **green**
- Action Button:
    - **Try Again**

This allows the user to replay and improve memory performance

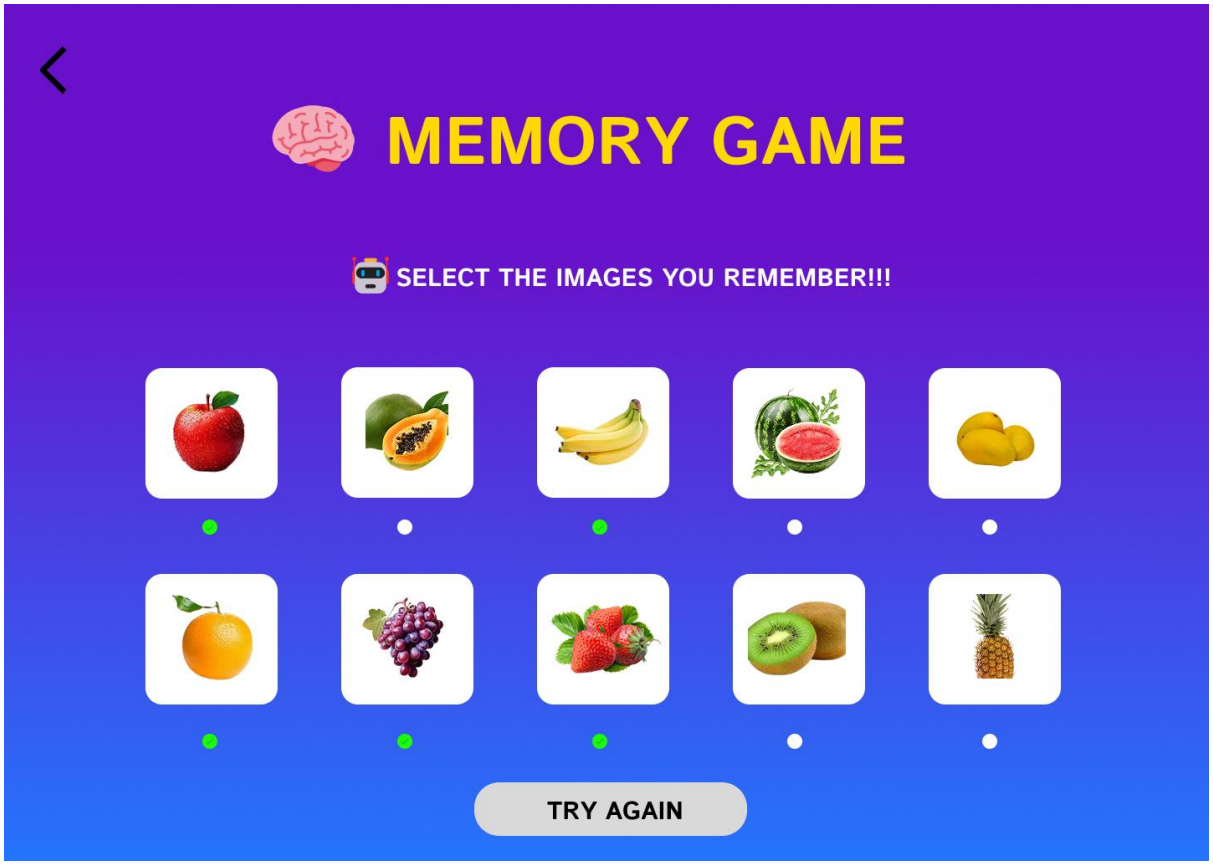
If User Clicks a Wrong Image

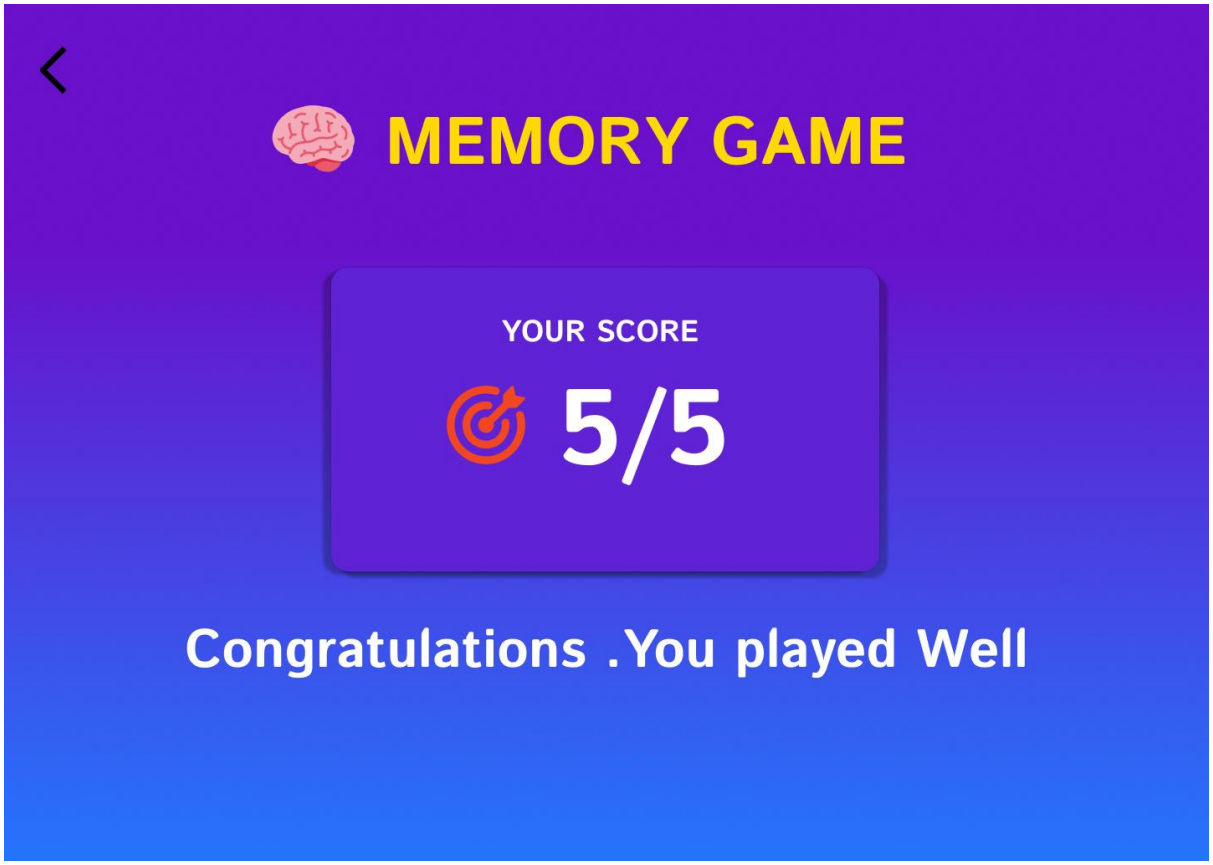
- Feedback Message:

“Sorry, you are wrong.”

- Visual Feedback:
    - Correct images → **Green**
    - Wrong image → **Red**
- Action Button:
    - **Try Again**

The user must restart the game.


MEMORY GAME
SELECT THE IMAGES YOU REMEMBER!!!
TRY AGAIN


MEMORY GAME
YOUR SCORE
5/5
Congratulations .You played Well

PROTOTYPE LINK:

https://www.figma.com/proto/jFJcWMQfy0jPyk8tPOTKLe/experiment2?node-id=7-20&t=Op1QO5aKs9dfLXBn-1&starting-point-node-id=7%3A20

Key Features of the Memory Game Design

1. Visual Memory-Based Gameplay

Users are shown a set of images and must recall them accurately, testing short-term memory and concentration.

2. Instant Selection Feedback

As soon as the user clicks an image:

- Correct choice → highlighted in **green**
- Wrong choice → highlighted in **red**

No submit button is required.

3. Real-Time Result Evaluation

- If **all selected images are correct**, the system shows **"Excellent Memory"**
- If **any wrong image is selected**, the system immediately shows **"Sorry, you are wrong"**

4. Retry-Based Interaction

After result display:

- Only **"Try Again"** option is shown
- User can replay without reloading the page

5. No Manual Submission

The game logic works automatically based on user interaction, making it faster and more engaging.